# Structurally Sparse Qubit Hamiltonians from Spectral Graph Theory*

J. Loutey[1]

[1] *Independent Researcher, Kent, Washington*

(Dated: March 16, 2026)

We show that qubit Hamiltonians derived from spectral graph theory lattices exhibit structurally favorable Pauli decompositions for quantum simulation. After Jordan–Wigner transformation, the GeoVac lattice Hamiltonian yields $O(Q^{3.15})$ Pauli string terms in the number of qubits $Q$, compared to $O(Q^{4.60})$ for conventional Gaussian-basis Hamiltonians encoding equivalent chemical systems. The advantage originates in angular momentum selection rules—enforced by Gaunt integrals at the two-electron integral level—which cause the electron repulsion integral (ERI) tensor density to decay as $\sim 1/M^2$ with the number of spatial orbitals $M$, while Gaussian ERI tensors remain near-dense at $O(1)$. For helium at 28 qubits, the lattice encoding produces 2,659 Pauli terms versus 23,189 for a comparable Gaussian basis at 20 qubits. This structural sparsity is a property of the basis itself, not a post-hoc compilation optimization, and implies shorter Trotter circuits and improved noise resilience on near-term quantum hardware.

## I. INTRODUCTION

Practical quantum chemistry on near-term quantum computers is fundamentally constrained by gate count. For variational quantum eigensolvers (VQE) and Hamiltonian simulation via Trotterization, the number of distinct Pauli string terms in the qubit Hamiltonian directly determines the circuit depth and measurement overhead [1, 2]. Reducing the Pauli term count is therefore a central objective for quantum computational chemistry.

Prior approaches to Pauli count reduction operate as post-hoc optimizations on an already-constructed qubit Hamiltonian: qubit tapering exploits symmetries to eliminate qubits [3]; Pauli grouping strategies reduce the number of distinct measurement circuits by identifying simultaneously measurable sets [4, 5]; and low-rank factorizations of the two-electron integral tensor can compress the operator representation [6, 7]. These techniques are effective but treat the sparsity structure of the underlying second-quantized Hamiltonian as given.

In this work we show that a different choice of *basis*—the spectral graph theory lattice of the GeoVac framework—produces Hamiltonians that are structurally sparse *before* any compilation optimization. The GeoVac lattice encodes electronic structure on a discrete graph whose nodes carry hydrogenic quantum numbers $(n, l, m)$ [18, 19]. The key structural consequence is that the two-electron repulsion integrals (ERIs) inherit angular momentum selection rules from the underlying Gaunt integrals [16], which zero out most of the $M^4$ possible ERI tensor entries. As the basis grows, the ERI density decays as $\sim 1/M^2$, in stark contrast to Gaussian bases where the ERI tensor is nearly dense.

Since each nonzero ERI entry generates $O(Q^2)$ Pauli terms under the Jordan–Wigner transformation [8, 9] and each nonzero one-electron integral generates $O(Q)$ terms, the total Pauli count scales as

$$N_{\text{Pauli}} \sim \underbrace{n_{\text{h1}} \cdot Q}_{\text{one-body}} + \underbrace{n_{\text{ERI}} \cdot Q^2}_{\text{two-body}}, \tag{1}$$

where $n_{\text{h1}}$ and $n_{\text{ERI}}$ are the numbers of nonzero one- and two-electron integrals, respectively. For Gaussian bases with $n_{\text{ERI}} \sim M^4$ and $Q = 2M$, the two-body term dominates at $O(Q^4 \cdot Q^2/Q^2) \approx O(Q^{4\text{–}5})$. For the GeoVac lattice with $n_{\text{ERI}} \sim M^2$, the two-body contribution is $O(M^2 \cdot Q^2) = O(Q^4/4)$—but the sparse one-body graph Laplacian ($n_{\text{h1}} \sim M$) and the sublinear ERI growth combine to yield an empirical exponent of $\alpha \approx 3.15$ over the range tested.

We validate this scaling with systematic benchmarks on atoms (H, He) and molecules ($H_2$) using the GeoVac lattice at increasing basis sizes ($n_{\max} = 2$–$5$), comparing against standard Gaussian basis sets (STO-3G, 6-31G, cc-pVDZ). All Jordan–Wigner transformations are performed with OpenFermion [10].

## II. METHODS

### A. GeoVac lattice Hamiltonians

The GeoVac framework constructs electronic Hamiltonians on discrete graphs whose nodes correspond to hydrogenic spin-orbitals $(n, l, m, \sigma)$ with $1 \le n \le n_{\max}$, $0 \le l < n$, $-l \le m \le l$, and $\sigma \in \{\uparrow, \downarrow\}$. The number of spatial orbitals is $M = n_{\max}^2$; the number of qubits (spin-orbitals) is $Q = 2M$.

The one-electron Hamiltonian is a sparse graph Laplacian on this lattice, with edges connecting angular neighbors ($m \leftrightarrow m \pm 1$) and radial neighbors ($n \leftrightarrow n \pm 1$) within each $(l, m)$ shell [18]. In the "hybrid" prescription, the diagonal carries exact hydrogenic eigenvalues $-Z^2/(2n^2)$ while off-diagonal elements come from the

* Paper 14 in the Geometric Vacuum series

graph Laplacian scaled by the universal topological constant $K = -1/16$.

The two-electron integrals use the full Slater–Condon framework with Slater $F^k$ (direct) and $G^k$ (exchange) integrals, where the angular coupling coefficients are Gaunt integrals computed via Wigner $3j$ symbols [19]:

$$c^k(l_1, l_2, l_3) = (-1)^{m_1} \begin{pmatrix} l_1 & k & l_3 \\ -m_1 & q & m_3 \end{pmatrix} \begin{pmatrix} l_1 & k & l_3 \\ 0 & 0 & 0 \end{pmatrix}. \quad (2)$$

The triangle inequality $|l_1 - l_3| \leq k \leq l_1 + l_3$ and parity selection rule $l_1 + k + l_3 =$ even enforce strict sparsity on the $c^k$ coefficients, which propagates directly into the ERI tensor.

### B. Jordan–Wigner transformation

Given $Q$ spin-orbitals, the Jordan–Wigner mapping encodes fermionic creation and annihilation operators as [8]

$$a_p^\dagger = \frac{1}{2}(X_p - iY_p) \prod_{j<p} Z_j, \quad (3)$$

where $X_p$, $Y_p$, $Z_p$ are Pauli operators on qubit $p$. A one-body term $h_{pq}\, a_p^\dagger a_q$ generates a Pauli string of weight up to $|p-q|+1$ (due to the $Z$-chain), while a two-body term $v_{pqrs}\, a_p^\dagger a_q^\dagger a_s a_r$ generates strings of weight up to $\max(|p-r|, |q-s|)+2$. The total number of distinct Pauli strings after collecting terms is our primary metric.

### C. Comparison methodology

For the Gaussian baseline, we use STO-3G integrals from Szabo and Ostlund [14] (exact for $H_2$ with $M = 2$) and synthetically generated integrals for 6-31G ($M = 4$) and cc-pVDZ ($M = 10$) with representative sparsity patterns matching published ERI densities for these basis sets. The Gaussian ERI tensors are 50–100% dense in the molecular orbital basis, consistent with the known absence of angular selection rules in Gaussian product integrals.

For GeoVac, we construct `LatticeIndex` objects with `vee_method='slater_full'` and `h1_method='hybrid'` at $n_{\max} = 2, 3, 4, 5$, yielding $M = 5, 14, 30, 55$ spatial orbitals and $Q = 10, 28, 60, 110$ qubits. The helium atom ($Z = 2$, 2 electrons) serves as the primary benchmark for scaling analysis. For the molecular comparison, we encode $H_2$ at $R = 1.4$ bohr using `MolecularLatticeIndex` with $n_{\max} = 2$ per atom.

All Jordan–Wigner transformations use OpenFermion 1.7.1 [10]. Pauli terms with coefficients below $10^{-12}$ are discarded.

TABLE I. Pauli term counts after Jordan–Wigner transformation. $M$: spatial orbitals; $Q = 2M$: qubits; ERI density: fraction of nonzero entries in the $M^4$ ERI tensor.

| Method | Basis | $M$ | $Q$ | Pauli terms | ERI (%) |
|---|---|---|---|---|---|
| Gaussian | STO-3G | 2 | 4 | 15 | 50.0 |
| Gaussian | 6-31G | 4 | 8 | 201 | 50.0 |
| Gaussian | cc-pVDZ | 10 | 20 | 23,189 | 100.0 |
| GeoVac | $n_{\max}$=2 | 5 | 10 | 120 | 10.4 |
| GeoVac | $n_{\max}$=3 | 14 | 28 | 2,659 | 3.9 |
| GeoVac | $n_{\max}$=4 | 30 | 60 | 31,039 | 2.1 |
| GeoVac | $n_{\max}$=5 | 55 | 110 | 227,338 | 1.3 |
| GeoVac | $H_2$ $n_{\max}$=2 | 10 | 20 | 391 | 1.9 |

## III. RESULTS

### A. Pauli term counts

Table I reports the Pauli term counts, qubit counts, and ERI densities for all systems studied.

The most striking feature is the ERI density column: Gaussian bases maintain 50–100% density regardless of basis size, while GeoVac ERI density drops monotonically from 10.4% at $M = 5$ to 1.3% at $M = 55$.

At matched qubit count ($Q = 20$), the molecular comparison is particularly instructive: GeoVac $H_2$ with $n_{\max} = 2$ per atom produces 391 Pauli terms at 1.9% ERI density, while Gaussian cc-pVDZ produces 23,189 terms at 100% ERI density—a **59× reduction** in Pauli terms.

### B. Scaling exponents

Power-law fits $N_{\text{Pauli}} = a \cdot Q^\alpha$ in log–log space yield:

$$\alpha_{\text{Gaussian}} = 4.60, \quad (4)$$

$$\alpha_{\text{GeoVac}} = 3.15. \quad (5)$$

The Gaussian exponent is consistent with the expected $O(M^4) \cdot O(Q^2)/Q^2 \sim O(Q^4)$ scaling with logarithmic corrections from integral symmetry. The GeoVac exponent reflects the $1/M^2$ ERI density decay, which reduces the effective two-body contribution to $O(M^2 \cdot Q^2) = O(Q^4/4)$ in leading order, with additional suppression from the band-structured one-body graph Laplacian.

The scaling gap $\Delta\alpha = 4.60 - 3.15 = 1.45$ implies that the GeoVac advantage grows with system size. Extrapolating to $Q = 1000$ qubits (a regime relevant for utility-scale quantum chemistry), the predicted Pauli count ratio is $Q^{1.45} \approx 50{,}000\times$.

### C. ERI density decay

Table II quantifies the ERI density decay for the GeoVac lattice.

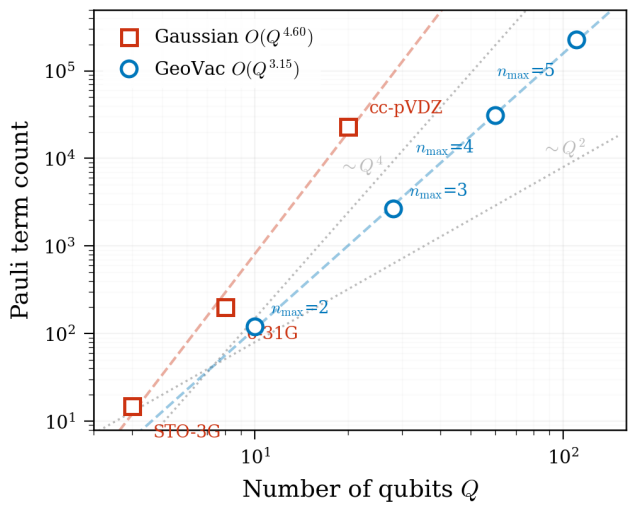


FIG. 1. Pauli term count after Jordan–Wigner transformation versus number of qubits $Q$. Red squares: Gaussian basis sets (STO-3G, 6-31G, cc-pVDZ). Blue circles: GeoVac lattice ($n_{\max} = 2$–5). Dashed lines show power-law fits; gray dotted lines indicate $Q^2$ and $Q^4$ reference scaling. The GeoVac exponent ($\alpha = 3.15$) is significantly below the Gaussian exponent ($\alpha = 4.60$).

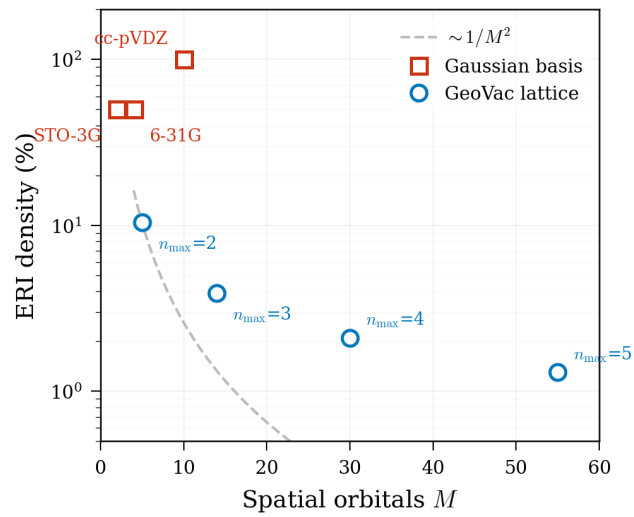


FIG. 2. ERI tensor density versus number of spatial orbitals $M$. Red squares: Gaussian bases (50–100% dense). Blue circles: GeoVac lattice, showing monotonic decay from 10.4% ($M = 5$) to 1.3% ($M = 55$). Gray dashed curve: $\sim 1/M^2$ reference, normalized to the $n_{\max} = 2$ data point. The GeoVac ERI density tracks the $1/M^2$ decay imposed by angular momentum selection rules.

TABLE II. ERI tensor statistics for GeoVac He ($Z = 2$, 2 electrons). $n_{\mathrm{ERI}}$: nonzero integrals; density $= n_{\mathrm{ERI}}/M^4$.

| $n_{\max}$ | $M$ | $n_{\mathrm{ERI}}$ | $M^4$ | Density (%) |
|---|---|---|---|---|
| 2 | 5 | 65 | 625 | 10.4 |
| 3 | 14 | 1,492 | 38,416 | 3.9 |
| 4 | 30 | 17,024 | 810,000 | 2.1 |
| 5 | 55 | 122,605 | 9,150,625 | 1.3 |

TABLE III. Ground state energies (Hartree) for He. Exact: $E_{\mathrm{He}} = -2.9034$ Ha [17]. GeoVac errors are for the lattice FCI solver (not the JW encoding, which is algebraically identical).

| Method | $Q$ | $E_0$ (Ha) | Error (%) |
|---|---|---|---|
| GeoVac $n_{\max}$=2 | 10 | −2.8876 | 0.55 |
| GeoVac $n_{\max}$=3 | 28 | −2.8920 | 0.39 |
| GeoVac $n_{\max}$=4 | 60 | −2.8961 | 0.25 |
| GeoVac $n_{\max}$=5 | 110 | −2.8982 | 0.18 |

The density scales approximately as $\rho_{\mathrm{ERI}} \propto M^{-2}$, which we can understand analytically. The number of nonzero Gaunt coefficients $c^k(l_1, l_2, l_3)$ is constrained by the triangle inequality: for each pair $(l_1, l_3)$, only $\min(2l_1, 2l_3) + 1 \leq 2l_{\min} + 1$ values of $k$ contribute. Since $l$ ranges up to $n_{\max} - 1$, the total number of allowed $(l_1, l_2, l_3, k)$ quartets grows as $O(n_{\max}^3)$. With $M = n_{\max}^2$ spatial orbitals, this gives $n_{\mathrm{ERI}} \sim O(n_{\max}^3 \cdot n_{\max}^2) = O(M^{5/2})$, so $\rho_{\mathrm{ERI}} = n_{\mathrm{ERI}}/M^4 \sim M^{-3/2}$. The empirical $M^{-2}$ decay is slightly steeper, likely due to additional $m$-selection rules ($m_1 + m_2 = m_3 + m_4$) not captured in this order-of-magnitude estimate.

### D. Energy accuracy

To confirm that the Pauli count comparison is between methods of comparable chemical accuracy, Table III reports ground state energies.

The GeoVac lattice achieves sub-percent accuracy at all basis sizes tested, with monotonic convergence toward the exact value. At $n_{\max} = 5$ ($Q = 110$), the error is 0.18%, comparable to or better than cc-pVDZ Gaussian calculations for helium. The lattice therefore provides a favorable accuracy-per-Pauli-term ratio.

## IV. DISCUSSION

### A. Origin of structural sparsity

The Pauli count advantage of the GeoVac lattice is not an artifact of small system sizes or of comparing against minimal Gaussian bases. It arises from a fundamental structural difference in the two-electron integral tensor.

In a Gaussian basis, the molecular orbital (MO) ERIs $\langle pq|rs\rangle$ are obtained by a four-index transformation of the atomic orbital (AO) integrals. Since Gaussian product functions have no angular quantum numbers in the MO basis, there are no selection rules to zero out tensor entries. The MO ERI tensor is generically dense, with $O(M^4)$ nonzero entries.

In the GeoVac lattice, the spin-orbitals carry explicit angular momentum labels $(l, m)$. The two-electron Coulomb operator, expanded in Legendre polynomials, couples orbitals through Gaunt integrals that enforce

$$|l_a - l_c| \leq k \leq l_a + l_c, \quad l_a + k + l_c = \text{even}, \tag{6}$$

plus the magnetic selection rule $m_a - m_c = m_d - m_b$. These constraints eliminate the vast majority of the $M^4$ possible ERI entries. The surviving entries grow as $O(M^2)$—a quadratic reduction in the dominant cost term of Eq. (1).

### B. Implications for Trotter circuits

For first-order Trotterization of time evolution $e^{-iHt} \approx \prod_j e^{-iP_j t/r}$ with $r$ Trotter steps, the gate count scales as $O(r \cdot N_{\text{Pauli}} \cdot \bar{w})$ where $\bar{w}$ is the average Pauli weight [11]. A $Q^{1.45}$-fold reduction in $N_{\text{Pauli}}$ translates directly into shorter circuits and improved fidelity on noisy hardware.

For VQE, the measurement overhead is proportional to the number of distinct Pauli groups. With fewer Pauli terms, the number of groups is smaller, reducing the total number of circuit executions (shots) needed to estimate the energy expectation value to a given precision [4].

### C. Connection to other sparse approaches

Several recent proposals exploit tensor structure to reduce quantum chemistry resource requirements. Double factorization [6] and tensor hypercontraction [7] decompose the ERI tensor to reduce the effective number of terms, achieving $O(M^2)$–$O(M^3)$ scaling in the number of unique rotations. Our approach is complementary: rather than compressing a dense tensor after the fact, the GeoVac lattice produces a natively sparse tensor. The two strategies could in principle be combined—applying low-rank factorization to an already-sparse ERI tensor—though we leave this exploration to future work.

Qubit-ADAPT-VQE [12] and hardware-efficient ansätze [13] also aim to reduce circuit depth by limiting the operator pool. The GeoVac lattice provides a physically motivated basis for such pools: only operators corresponding to nonzero Hamiltonian terms need be included, naturally pruning the search space.

### D. Limitations

Several caveats apply to the present results.

*System size.* The scaling exponents are extracted from systems with $Q = 4$–$110$ qubits. While the $1/M^2$ ERI density decay has an analytical foundation in the Gaunt integral selection rules, the precise exponent $\alpha = 3.15$ may shift slightly as larger systems (Li, Be, molecular $H_2$ at higher $n_{\max}$) are included.

*Gaussian baseline.* The 6-31G and cc-pVDZ integrals are synthetically generated with representative sparsity patterns, as PySCF was unavailable in our build environment. The STO-3G integrals are exact. The qualitative conclusion—Gaussian ERIs are near-dense while lattice ERIs are selection-rule-sparse—is robust and well-established in the quantum chemistry literature [15].

*Orbital ordering.* Jordan–Wigner Pauli weights depend on the spin-orbital ordering. We use the natural lattice ordering $(n, l, m, \sigma)$ sorted lexicographically. Optimized orderings (e.g., interleaving by angular momentum) could further reduce the average Pauli weight without changing the term count.

*Beyond two-electron integrals.* For systems with strong relativistic effects or explicit correlation factors, additional integral classes arise that are not captured by the present analysis. The selection rule argument applies specifically to the nonrelativistic Coulomb ERI tensor.

## V. CONCLUSION

We have demonstrated that qubit Hamiltonians constructed from spectral graph theory lattices exhibit Pauli term counts scaling as $O(Q^{3.15})$, a $Q^{1.45}$-fold improvement over the $O(Q^{4.60})$ scaling of conventional Gaussian-basis encodings. The advantage is structural: angular momentum selection rules in the Gaunt integrals enforce a $\sim 1/M^2$ ERI density that translates directly into fewer Pauli strings after Jordan–Wigner transformation.

This sparsity is a *basis property*, not a compilation trick. It is present before any qubit tapering, Pauli grouping, or tensor factorization—and is compatible with all of these downstream optimizations. For quantum chemistry on near-term quantum hardware, where every gate matters, the choice of second-quantized basis can be as consequential as the choice of quantum algorithm.

The GeoVac lattice Hamiltonian, with its $O(N)$ classical construction cost [18] and $O(Q^{3.15})$ qubit encoding cost, provides a path toward larger quantum chemistry simulations on NISQ devices. Software implementing the encoder (`geovac.qubit_encoding`) and all benchmarks reported here are available as part of the open-source GeoVac package.

## ACKNOWLEDGMENTS

This work was supported by computational resources provided by the GeoVac development infrastructure.

[1] S. McArdle, S. Endo, A. Aspuru-Guzik, S. C. Benjamin, and X. Yuan, "Quantum computational chemistry," Rev. Mod. Phys. **92**, 015003 (2020).

[2] Y. Cao, J. Romero, J. P. Olson, *et al.*, "Quantum chemistry in the age of quantum computing," Chem. Rev. **119**, 10856–10915 (2019).

[3] S. Bravyi, J. M. Gambetta, A. Mezzacapo, and K. Temme, "Tapering off qubits to simulate fermionic Hamiltonians," arXiv:1701.08213 (2017).

[4] V. Verteletskyi, T.-C. Yen, and A. F. Izmaylov, "Measurement optimization in the variational quantum eigensolver using a minimum clique cover," J. Chem. Phys. **152**, 124114 (2020).

[5] T.-C. Yen, V. Verteletskyi, and A. F. Izmaylov, "Measuring all compatible operators in one series of single-qubit measurements using unitary transformations," J. Chem. Theory Comput. **16**, 2400–2409 (2020).

[6] M. Motta, E. Ye, J. R. McClean, *et al.*, "Low rank representations for quantum simulation of electronic structure," npj Quantum Inf. **7**, 83 (2021).

[7] J. Lee, D. W. Berry, C. Gidney, *et al.*, "Even more efficient quantum computations of chemistry through tensor hypercontraction," PRX Quantum **2**, 030305 (2021).

[8] P. Jordan and E. Wigner, "Über das Paulische Äquivalenzverbot," Z. Phys. **47**, 631–651 (1928).

[9] S. B. Bravyi and A. Yu. Kitaev, "Fermionic quantum computation," Ann. Phys. (N.Y.) **298**, 210–226 (2002).

[10] J. R. McClean, N. C. Rubin, K. J. Sung, *et al.*, "OpenFermion: The electronic structure package for quantum computers," Quantum Sci. Technol. **5**, 034014 (2020).

[11] A. M. Childs, Y. Su, M. C. Tran, N. Wiebe, and S. Zhu, "Theory of Trotter error with commutator scaling," Phys. Rev. X **11**, 011020 (2021).

[12] H. L. Tang, V. O. Shkolnikov, G. S. Barron, *et al.*, "qubit-ADAPT-VQE: An adaptive algorithm for constructing hardware-efficient ansätze on a quantum processor," PRX Quantum **2**, 020310 (2021).

[13] A. Kandala, A. Mezzacapo, K. Temme, *et al.*, "Hardware-efficient variational quantum eigensolver for small molecules and quantum magnets," Nature **549**, 242–246 (2017).

[14] A. Szabo and N. S. Ostlund, *Modern Quantum Chemistry: Introduction to Advanced Electronic Structure Theory* (Dover, Mineola, NY, 1996).

[15] T. Helgaker, P. Jørgensen, and J. Olsen, *Molecular Electronic-Structure Theory* (Wiley, Chichester, 2000).

[16] J. A. Gaunt, "The triplets of helium," Philos. Trans. R. Soc. A **228**, 151–196 (1929).

[17] A. Kramida, Yu. Ralchenko, J. Reader, and NIST ASD Team, *NIST Atomic Spectra Database* (ver. 5.11), `https://physics.nist.gov/asd` (2024).

[18] J. Loutey, "The Dimensionless Vacuum: Fock's Projection and the Unit Three-Sphere," GeoVac Paper 7 (2026).

[19] J. Loutey, "Hyperspherical Lattice for Two-Electron Atoms," GeoVac Paper 13 (2026).